

Public Economics and Policy Lecture #3

Theory and Application of Public Finance II (The IS-LM Model)

18th August 2026

1

1

Learning objective

- ☐ Understand the relationship between fiscal policy and interest rates and be able to explain the '***crowding out***' effect.
- ☐ Understand the relationship between fiscal policy, interest rates, and exchange rates in an open economy.
- ☐ Apply an extended conceptual framework—combining equilibrium in the goods market with equilibrium in the money market—to analyze the '***effectiveness of fiscal policy***' in both closed and open economies.

2

Outline

- ❑ Extension framework of fiscal policy: equilibrium in good market and money market
 - ❑ IS-LM model in a closed economy
 - ❑ Crowding-out effect
 - ❑ IS-LM model in an open economy
- ❑ Review exercises

3

Reference

Reading:

- Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

Additional reading:

- Krish Bhaskar, & David F. Murray. (2015). *Macroeconomic Systems*. Routledge. (Chapter 4, 5)
- Thomas R. Michl. (2015). *Macroeconomic Theory: A Short Course*. Routledge. (Chapter 13, 14)

4

In the last lecture, ...

- ❑ We only consider the equilibrium in the output market.
 - ❑ **The change in government spending leads to the change in output (GDP).**
 - ❑ **Fiscal multiplier:**
 - ❑ **Expansionary fiscal policy: Govt spending $\uparrow$ $\rightarrow$ national income $\uparrow$**
(Contractionary fiscal policy: Govt spending $\downarrow$ $\rightarrow$ national income $\downarrow$)
 - ❑ **Tax multiplier:**
 - ❑ **Expansionary fiscal policy: Tax rate $\downarrow$ $\rightarrow$ national income $\uparrow$**
(Contractionary fiscal policy: Tax rate $\uparrow$ $\rightarrow$ national income $\downarrow$)

5

- ❑ But, in an economy, there exists both the output market and money market. $\rightarrow$ ***The change in government spending has an impact on both the output and the interest rate.***
- ❑ So, we need a model to study the equilibrium in the output market together with the equilibrium in the money market. $\rightarrow$ ***the IS-LM model***

6

Extension framework of Fiscal policy: Equilibrium in good market and money market

7

The IS/LM Analysis (in a closed economy)

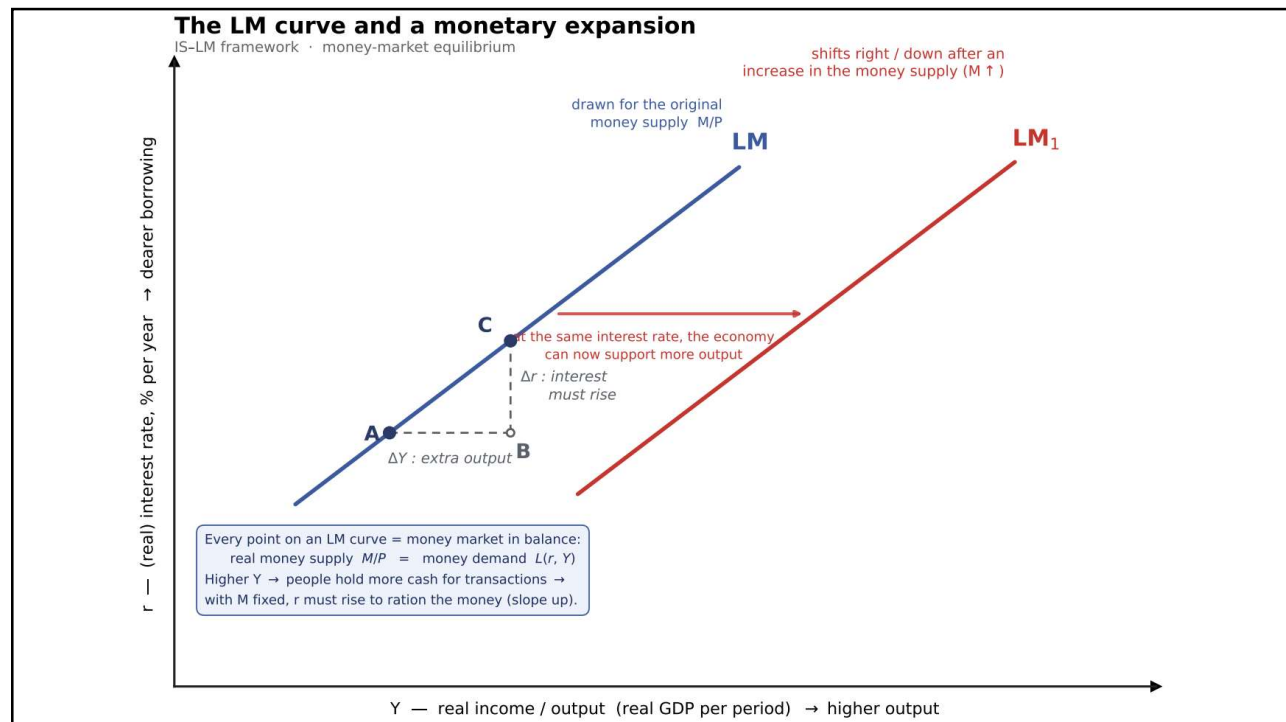
1 The Investment/Saving and Liquidity Preference/Money Supply Equilibrium Model

- ❑ Equilibrium in good market and equilibrium in money market
(see the lecture notes)

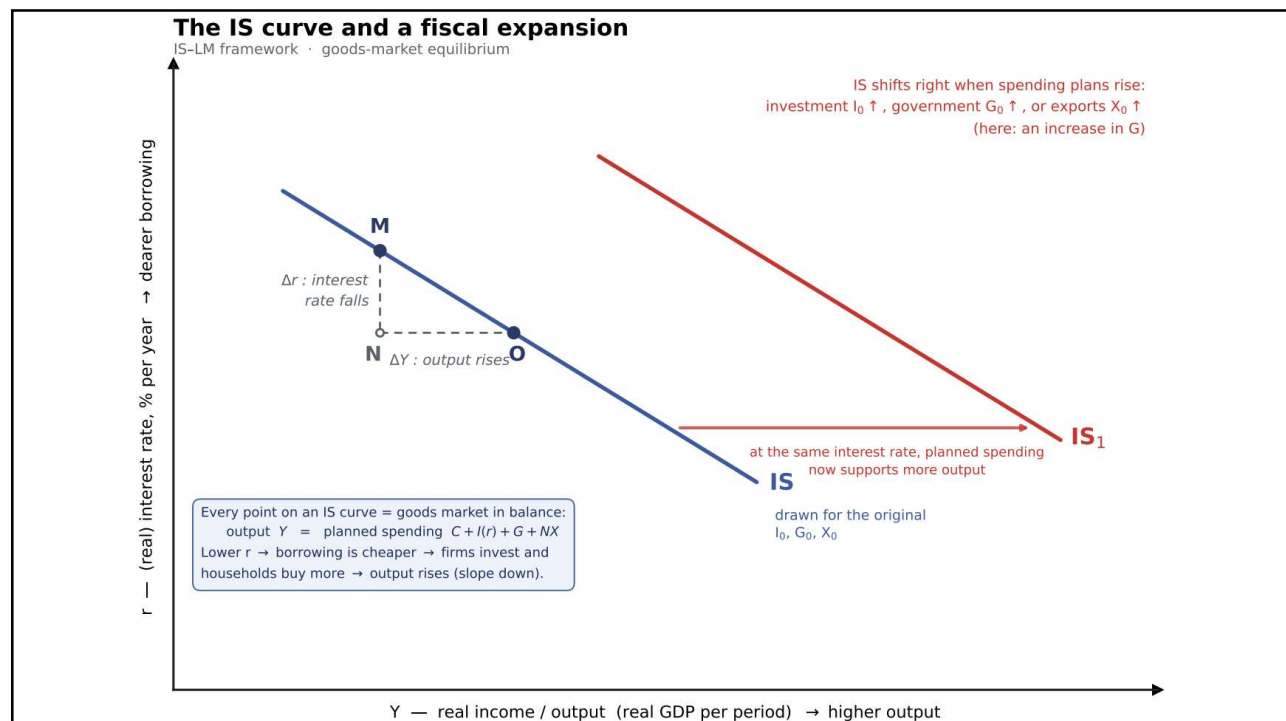
Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

8

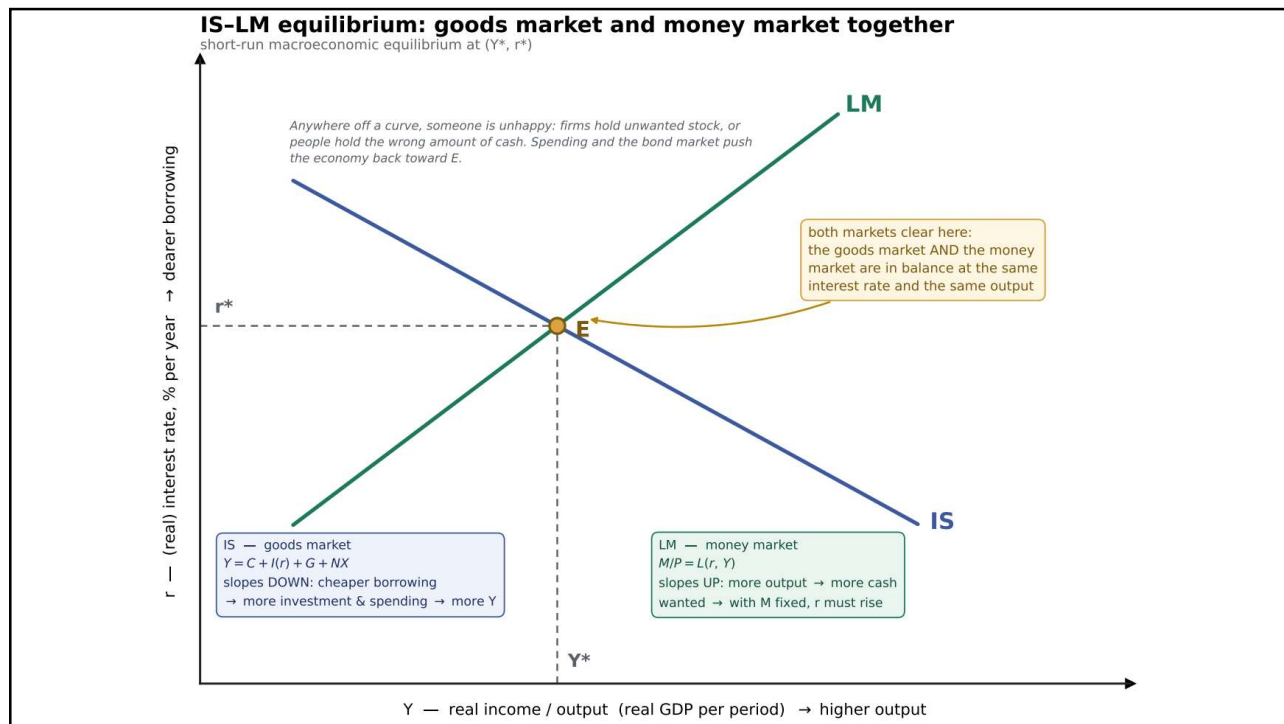
8



9



10



11

The IS/LM Analysis (in a closed economy)

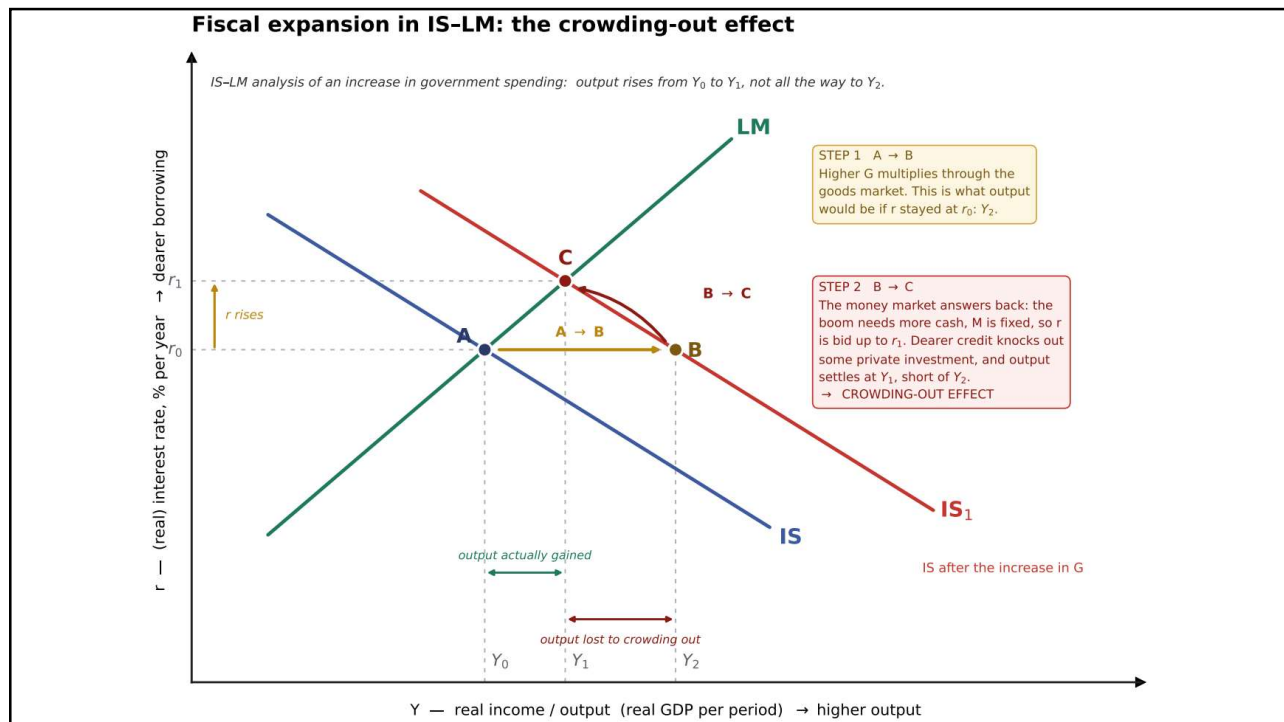
2 The Size of the Multiplier in the IS/LM Model

❑ The crowding-out effect

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

12

12



13

The IS/LM Analysis (in a closed economy)

3 The Crowding-In Effect

- ☐ If a public investment project makes private firms more productive, the **crowding-in effect** occurs.
- ☐ Fiscal policy may crowd in private investment or consumption, resulting in positive externalities in some cases.
- ☐ The crowding in effect can offset or reverse the crowding out effect.

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

14

14

Example: Crowding-in Effect

- ❑ Example: Public investment (infrastructure and utilities) in EEC (The Eastern Economic Corridor) attracts many firms to invest in as they believe in the prosperous growth and high yield of investment in this economic zone + increase in the value chain of related industry
- ❑ The demand and output of other industries (that are indirectly involved or related to the construction industry) will increase. For example,
 - ❑ Manufacturing industries, which supply the raw materials and equipment for the construction industry, such as steel, cement, machinery, and tools.
 - ❑ Transportation and logistics industries, which facilitate the movement and delivery of the materials and equipment for the construction industry, such as trucks, trains, ships, and planes.
 - ❑ The service industries, which provide various support and complementary services for the construction industry, such as engineering, design, consulting, finance, insurance, and maintenance.

15

Open economy

16

The Open Economy

1. Extension to an Open Economy Model

- Equilibrium condition in the goods market is

$$Y = C(Y) + I(r) + G + NX(e, Y)$$

- Equilibrium condition in the money market is

$$M = L(Y, r)$$

- When the world capital market is perfect, the rate of interest in the home country r is equal to the world rate of interest, r^* , so that the market for capital is in equilibrium. (If not, there will be capital inflow/outflow.)

$$r = r^*$$

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

17

17

Exchange rate system

Fixed exchange rate

- The value of a currency is pegged or fixed to the value of another currency or a basket of currencies, or to a commodity such as gold
- The central bank intervenes in the foreign exchange market to maintain the exchange rate within a narrow band or at a specific level.
- Advantages: Reduce uncertainty and volatility → promote trade and investment, and maintain price stability
- Disadvantages: Limit the monetary policy autonomy, create fiscal and external imbalances, and expose the economy to speculative attacks

Flexible exchange rate

- The value of a currency is determined by the supply and demand of the currency in the foreign exchange market
- The central bank does not intervene or minimally intervenes in the foreign exchange market to influence the exchange rate.
- Advantages: Enhance the monetary policy autonomy, adjust to external shocks, and correct fiscal and external imbalances
- Disadvantages: Increase uncertainty and volatility, discourage trade and investment, and generate inflationary or deflationary pressures → Firms can buy a hedged premium to cover their exchange rate risk (Other methods include forward contracts, futures contracts, options contracts and swaps)

18

Exchange rate system

Fixed exchange rate (e.g. Saudi Arabia)



Managed floating exchange rate (e.g. Thailand)



19

The Open Economy

2. The Fixed Exchange System and perfect capital mobility

☐ Fiscal policy is effective under a fixed exchange system.

3. The Flexible Exchange Rate System and perfect capital mobility: The Effect of Fiscal Policy

☐ Fiscal policy is ineffective.

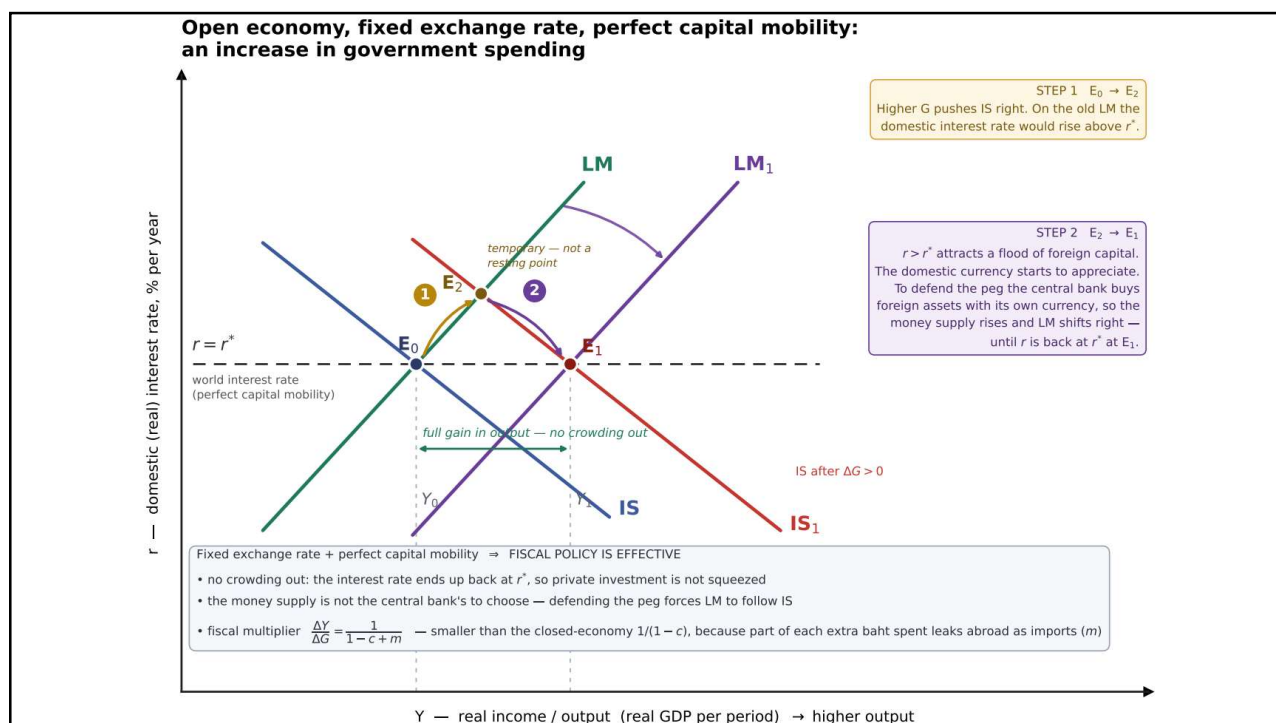
☐ Monetary policy is effective under a flexible exchange rate system.

☐ The impact of fiscal policy in a situation where capital mobility is not perfect is a combination of the perfect capital mobility example and the capital immobility example.

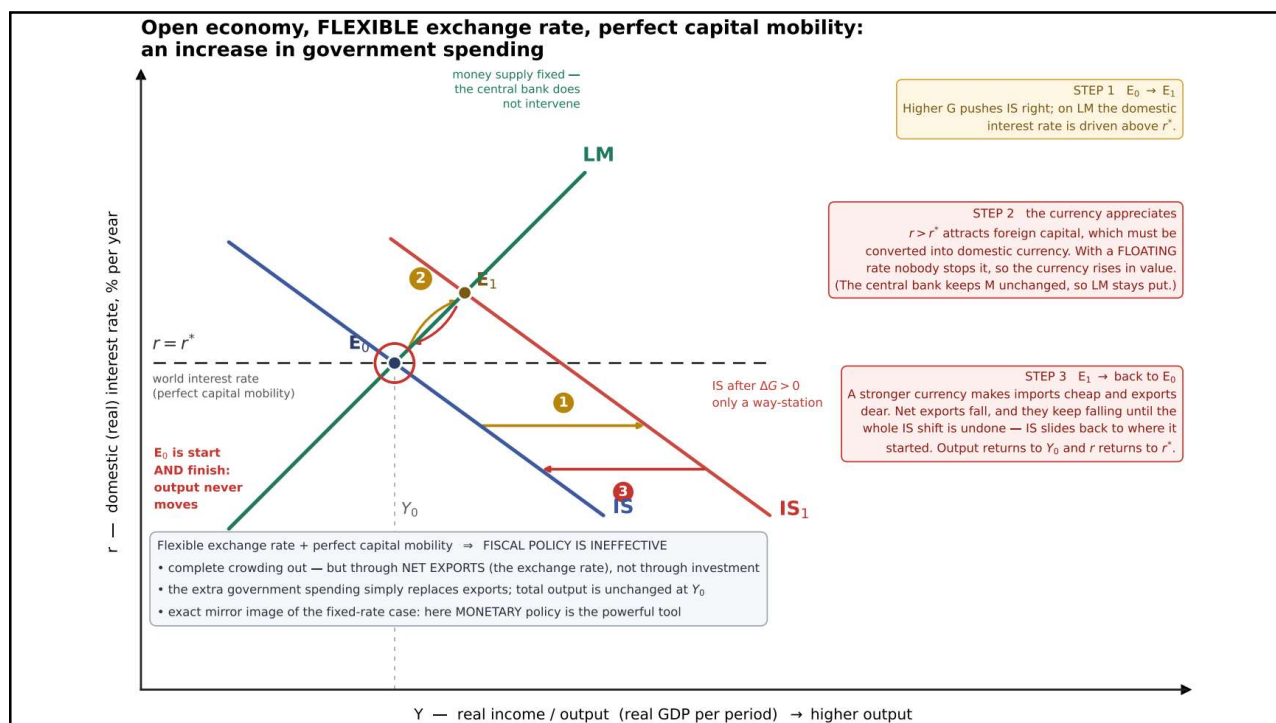
Source: Ihuri, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

20

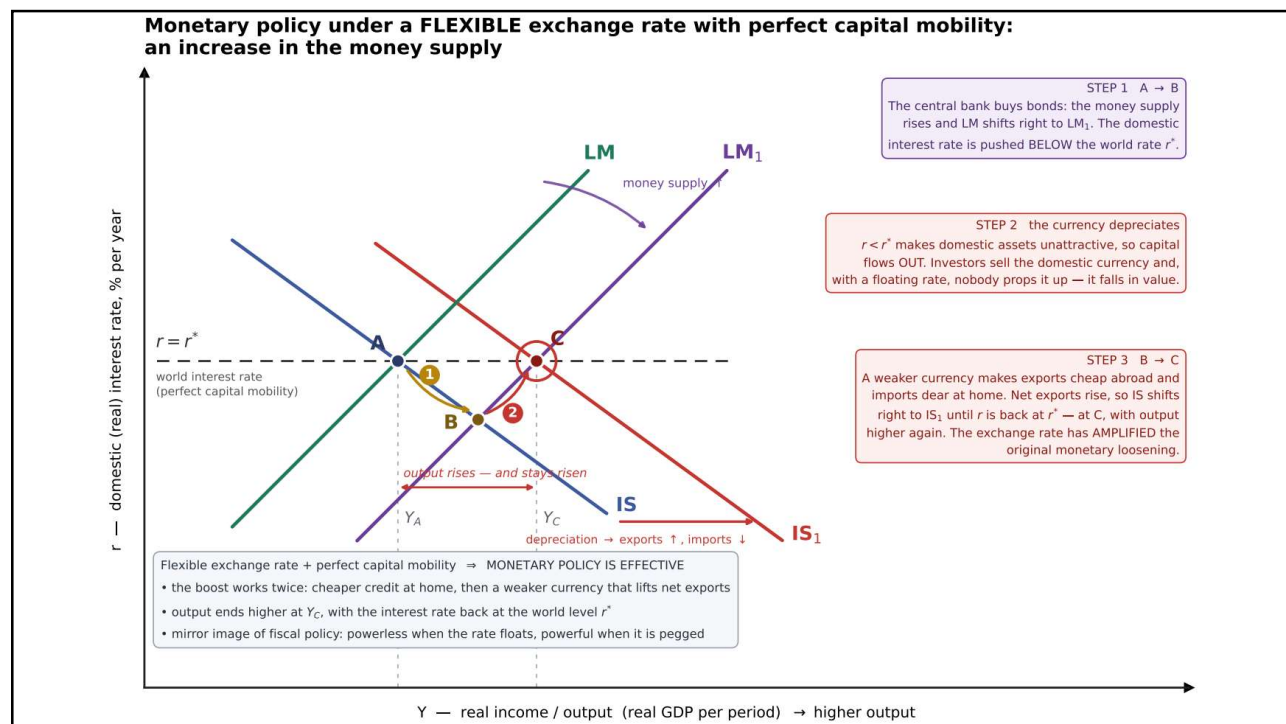
20



21



22



23

Conclusions

Fiscal policy, interest rates and the exchange rate — what the IS-LM framework tells us

1 Fiscal policy and interest rates in a CLOSED economy

- IS (goods market) and LM (money market) together pin down output and the interest rate — neither curve does it alone.
- A rise in G shifts IS right; the boom raises money demand and, with M fixed, the interest rate is bid up from r_0 to r_1 .
- Dearer credit depresses private investment, so output stops at Y_1 , short of the full-multiplier Y_2 — CROWDING OUT.
- How severe? Steep LM = severe; flat LM (liquidity trap) = almost none.

2 OPEN economy: the exchange rate decides the outcome

- Perfect capital mobility pins the domestic rate to the world rate r^* ; any gap triggers immediate capital flows.
- FIXED rate: defending the peg forces M to expand, so LM follows IS — no crowding out, fiscal policy effective. The multiplier $1/(1-c+m)$ is still damped by import leakage.
- FLOATING rate: the capital inflow appreciates the currency, net exports fall and IS slides all the way back — complete crowding out, via NX rather than investment.

3 Using the framework

- The channel of crowding out changes with the regime: private investment (closed) → none (fixed) → net exports (floating).
- Short run only: sticky prices, spare capacity, perfectly mobile capital. At full employment extra G becomes inflation.

IS-LM / Mundell-Fleming, short run with sticky prices

Effectiveness with perfect capital mobility

	Fixed rate	Floating rate
Fiscal policy	Powerful	Powerless
Monetary policy	Powerless	Powerful

The one idea to keep

Fiscal policy is never simply “effective” or “ineffective”.

- Its power depends on what the interest rate is free to do — and that is decided by the exchange-rate regime.
- Impossible trinity: free capital movement, a fixed exchange rate, an independent monetary policy — choose two.

Pegged → fiscal policy leads, money follows. Floating → monetary policy leads, fiscal policy is washed out.

24

Assigned Question (for Lecture 3)

1. Discuss why fiscal stimulus is expected to have a smaller impact on national income in the IS–LM model compared to the Keynesian model. (Long answer with a relevant diagram is expected.)
2. Discuss whether a fiscal stimulus would be effective in Malaysia. (Long answer with a relevant diagram is expected.)

25



26